Project Summary Report

Spin the Wheel!

Interactive Storytelling About Gender Identities Across Different Times and Places in Sweden

Group 01

Stakeholder: Umeå Kommun – Gender Equality Office

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1. Project Background

Understanding gender inequality requires not only awareness of present-day issues but also knowledge of historical and cultural contexts. Gender identities, social roles, and everyday life experiences have differed significantly across time periods and across regions in Sweden. The original *"Spin the Wheel!"* project demonstrated that interactive storytelling can be an effective way to create empathy and reflection by placing users in randomized identities across different historical times and locations.

However, the existing solution relies on manually created life scenarios, which limits scalability, diversity, and flexibility. This project proposes a generative AI-based interactive system that can automatically create historically and culturally grounded life stories and images. The generated content must be trustworthy, traceable, and unbiased, particularly because commercial AI models often reflect dominant narratives and introduce social and cultural bias.

2. Problem Description

The current *"Spin the Wheel!"* system depends on manually created scenarios, which limits the number of regions, time periods, and gender identities that can be represented. There is currently no AI-based system that can automatically generate historically accurate, culturally grounded, and traceable scenarios about gender identities in Sweden. At the same time, the naive use of generative AI introduces risks such as bias, stereotyping, and historical distortion. This creates a gap for educators and gender equality practitioners who need a scalable, transparent, and trustworthy interactive tool to support learning and reflection on gender inequality.

3. Research Focus and Expected Outcomes

Research Focus

The main research focus of this project is to explore:

> How a generative, interactive storytelling system can be designed to produce historically accurate, unbiased, and traceable gender identity scenarios across different time periods and regions in Sweden.

Expected Outcomes

- A conceptual architecture for an AI-based “Spin the Wheel!” system
- A scenario generation pipeline producing both text and images using traceable historical and cultural sources
- Alignment of generated scenarios with specific Swedish regions and time periods
- A gamified prototype interface that allows users to spin through and compare scenarios
- Basic explainability features showing how scenarios are generated
- An ethical and trustworthiness evaluation using ALTAI and related frameworks

4. Preliminary Design

The proposed system follows a layered architecture that separates user interaction, game logic, knowledge management, and AI-based content generation to ensure accuracy, transparency, and scalability.

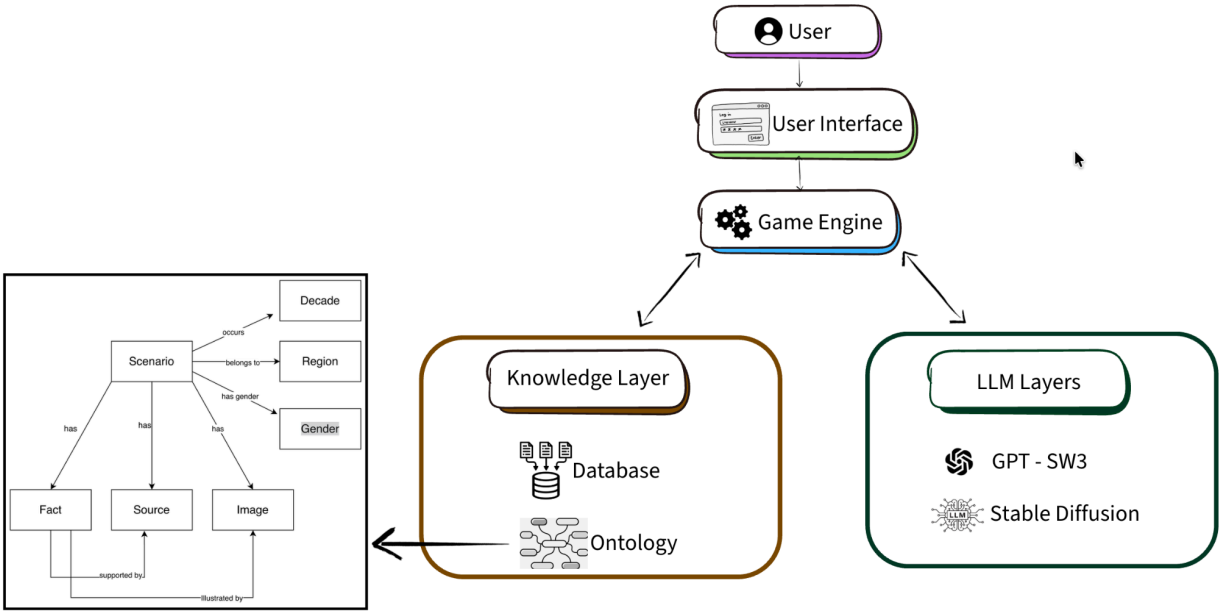

Figure 01 - Preliminary Design

The User interacts with the system through the User Interface (UI), which provides the visual and interactive elements of the game, such as spinning the wheel and viewing generated scenarios. The UI is

connected to the Game Engine, which controls the overall system flow. The Game Engine manages user actions, triggers scenario generation, and coordinates communication between the Knowledge Layer and the AI models.

The Knowledge Layer is the foundation for trustworthiness and historical accuracy. It consists of a database that stores verified historical facts, regions, decades, gender-related information, and sources, as well as an ontology that defines how these elements are connected. Each scenario is linked to a specific decade, region, and gender, and is supported by facts, sources, and images. This structured knowledge ensures that the generated content follows correct historical and cultural relationships.

The LLM Layer is responsible for content generation. GPT-SW3 is used to generate historically and culturally grounded text scenarios, while Stable Diffusion generates the corresponding visual images. The Game Engine sends controlled prompts to these models using constraints from the Knowledge Layer, which helps reduce bias and improve historical accuracy.

Finally, the generated scenarios are validated using the predefined rules and then presented to the user through the interface. This separation between structured knowledge and generative models supports bias control, explainability, and transparency, making the system suitable for trustworthy educational use.

5. ALTAI Evaluation (Trustworthy AI)

The project applies the ALTAI (Assessment List for Trustworthy Artificial Intelligence) framework to evaluate ethical and social responsibility:

- **Human Agency & Oversight:** Users freely explore scenarios, while experts can review generated outputs.
- **Transparency & Explainability:** Scenarios are linked to traceable sources and generation logic.
- **Fairness & Bias Mitigation:** Bias control mechanisms reduce stereotypes related to gender, region, and class.
- **Technical Robustness & Safety:** Harmful, misleading, or offensive content is filtered.
- **Privacy & Data Governance:** Only public, non-personal historical data is used.
- **Societal Well-being:** The system promotes awareness, education, and reflection on gender equality.

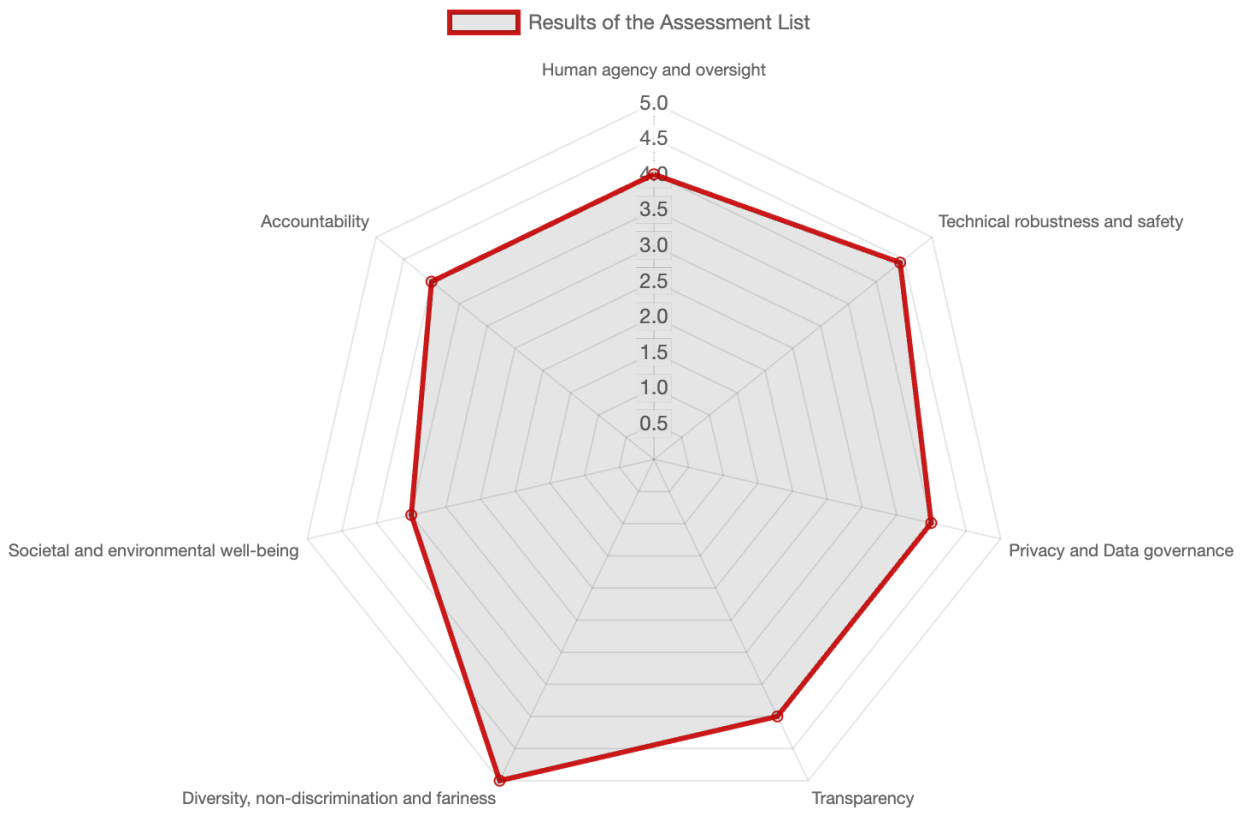

Figure 02 - ALTAI Assessment Result

6. Group Responsibilities and Task Distribution

Group Member Name	**Assigned Task**
Sérgio Costa	Preliminary system design & architecture Background research & problem formulation
Michail	Research question & expected outcomes
Chathumini Kavishya	ALTAI evaluation & ethical analysis Slide design & documentation